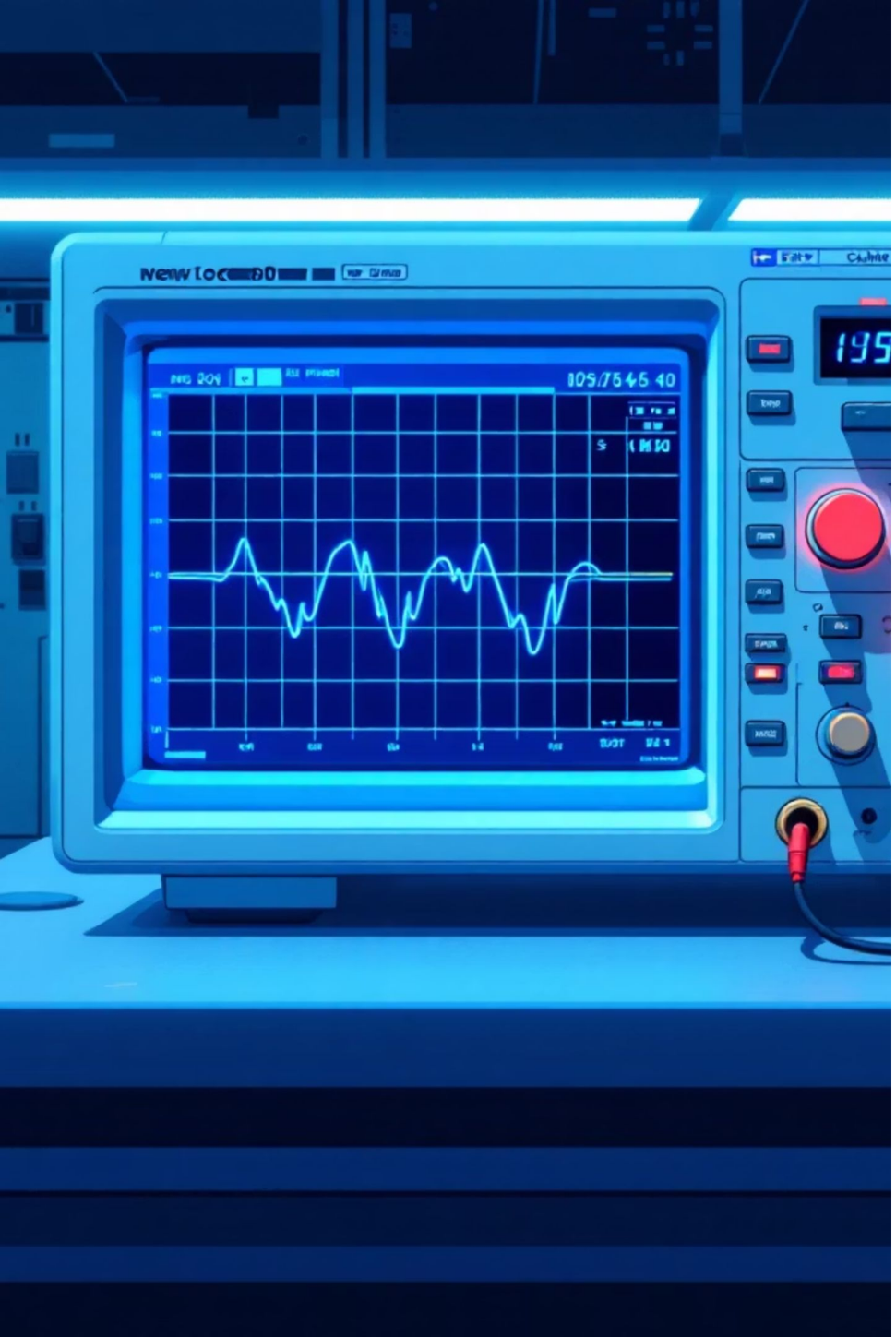


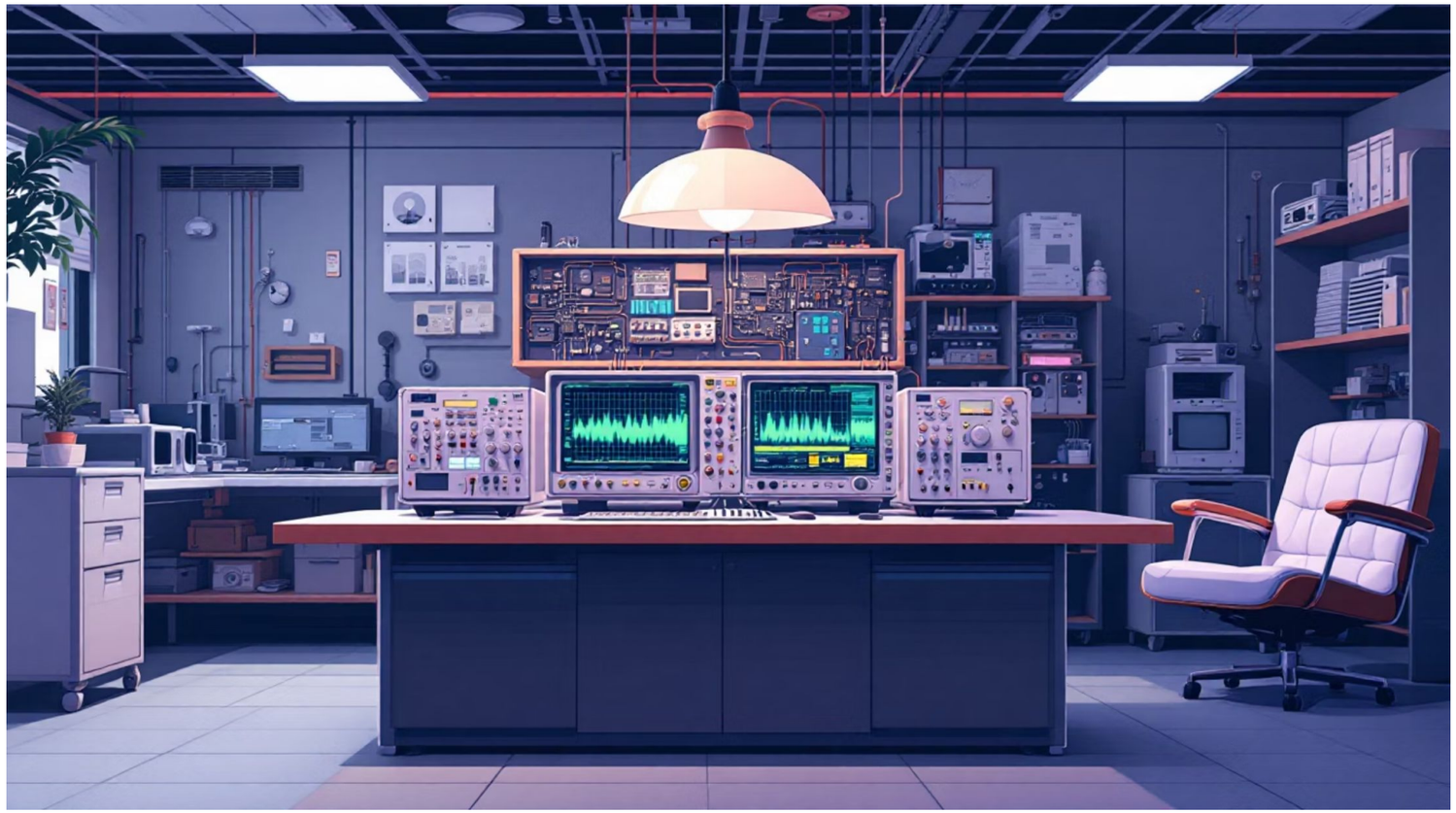
WiFi-CRO: AI-Powered Portable Oscilloscope

Replacing ₹50,000 lab equipment with an ₹800 intelligent measurement device

> Measure Anything. Understand Everything.

TEAM DRACARYS

The Problem: Lab Equipment is Out of Reach



The Reality

Traditional oscilloscopes cost between ₹30,000 and ₹2,00,000. They are bulky, lab-bound instruments that require specialized training to operate.
These devices display raw waveforms with no intelligent analysis. They show what the signal looks like but give no guidance on what the circuit is doing or whether it is healthy.

The Gap

Students and field engineers have no access to professional measurement tools. There is a massive gap between expensive lab equipment and nothing at all.

WiFi-CRO: Professional Measurement for ₹800

A portable WiFi oscilloscope built for ₹800 that streams live waveforms to any browser with full AI-powered analysis

±30V Range

Safe operation with ±51.8V absolute maximum

±30A Standard

Safe operation with ±20A

215Hz Bandwidth

430 SPS per channel

16-Bit ADC

62.5 microvolt resolution

Current Resolution	1mA minimum detectable change
Sample Modes	1000 / 2000 / 3000 / 4000 samples
Capture Window	2.3 to 9.3 seconds
Platform	Any browser — no software install

Technical Innovation: Hardware Architecture

Differential measurement architecture using ADS1115 16-bit ADC for full-cycle capture

01

Differential Mode Capture

ADS1115 in differential mode captures full positive AND negative cycles — no signal clipping

02

Voltage Scaling

Voltage divider (R1=147kΩ, R2=10kΩ) scales ±30V signals safely to ADS1115 input range

03

Current Sensing

ACS712 Hall effect sensor measures current non-invasively — no circuit cutting needed

04

WiFi Streaming

NodeMCU ESP8266 streams data over WiFi to browser in real time

05

High-Speed I2C

400kHz communication achieves 430 SPS per channel by alternating readings

Technical Innovation: FFT and THD Analysis

Fast Fourier Transform (FFT)

FFT decomposes complex waveforms into individual frequency components — like a prism splitting light into a rainbow. The device collects 1000 to 4000 samples, runs Cooley-Tukey FFT algorithm in browser JavaScript, producing live frequency spectra showing exactly which frequencies are present and how strong each one is.

Total Harmonic Distortion (THD)

THD measures signal purity — how much of the signal is unwanted harmonic distortion versus clean fundamental frequency.

$$THD\% = \frac{\sqrt{\sum \text{harmonic amplitudes}^2}}{\text{fundamental amplitude}} \times 100$$

Real world significance: Motors, power supplies, transformers have characteristic THD signatures. High THD indicates faults before failure occurs.

1000	2000	4000
Samples	**Samples**	**Samples**
0.43Hz resolution	0.22Hz resolution	0.11Hz resolution

Sample mode strategy: 1000 samples = zoom in, see waveform shape clearly. 4000 samples = maximum FFT accuracy, finest THD measurement.

5 Measurement Metrics: Comprehensive Analysis

Simultaneously calculates and displays these measurements live

Voltage Measurements

- Vdc (average)
- Vpeak (highest point)
- Vpp (peak to peak)
- Vrms (true RMS)
- Vcrest factor (pure sine=1.414)
- Vripple percentage

Current Measurements

- Idc, Ipeak, Ipp
- Irms
- Icrest factor
- Iform factor

Power Measurements

- Real power (watts)
- Apparent power (VA)
- Reactive power (VAR)
- Power factor (0 to 1)

Frequency Detection

Detected via zero crossing algorithm — upward crossings only for accuracy

Waveform Quality

FFT spectrum + THD percentage for both voltage and current

AI Integration: The Game Changer

Integrates GEMINI AI (gemini 2.5 flash) directly into the measurement workflow

Data and Waveform Image Collection

All metrics sent to Gemini AI API

Circuit Identification

Battery, charger, transformer, rectifier, or unknown

Health Diagnosis

Good ✅ Warning ⚠ Fault ❌

Health Score

0 to 100 with sub-scores

Trend Monitor

Tracks last 5 captures, warns of drifting values

Real World Significance

- Motor bearing faults show rising THD before physical failure — AI catches early
- Power factor below 0.8 triggers automatic warning

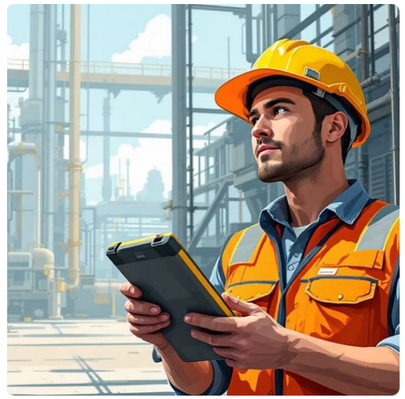
Impact

No oscilloscope expertise needed — AI explains what it sees in plain language

Real World Applications and Long Term Impact

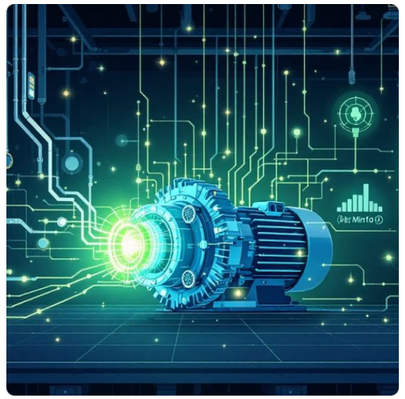
Education

Students visualize AC/DC circuits, transformers, rectifiers in real time. One device serves entire lab wirelessly.



Field Engineering

Technicians carry ₹800 device instead of ₹50,000 oscilloscope. Measure motor health, inverter quality, solar output on site.

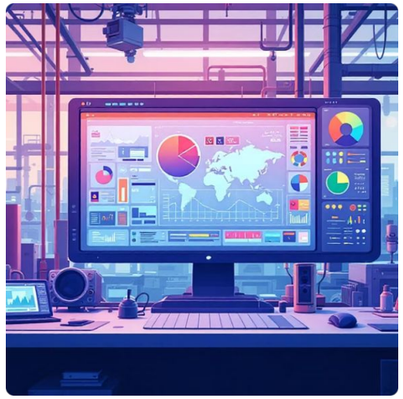


Preventive Maintenance

AI trend monitoring detects rising THD before physical failure — saves lakhs in repair costs.

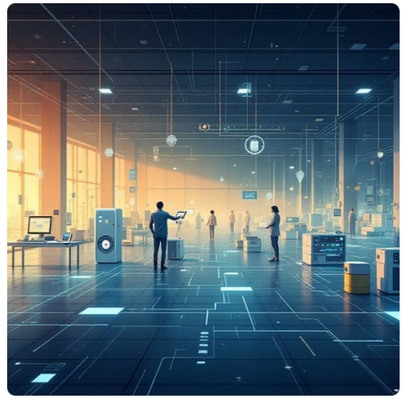
Power Quality

Measure harmonics in industrial supply — high THD causes motor heating and energy waste — identifies instantly.



Remote Monitoring

Multiple devices on same WiFi network — monitor different factory parts simultaneously from one laptop.



Industry 4.0 Vision

Intelligent portable devices become essential infrastructure — not just lab tools.

Future Development Roadmap

Near Term: Enhanced Range

Increase voltage and current range. Add MCP3202 SPI ADC (100k SPS) → extend frequency range to 50kHz covering full audio spectrum

Medium Term: Professional Design

Custom PCB replacing breadboard → professional form factor, reliable connections, compact size. Add OLED display for basic readings without browser

Field Ready

Waterproof enclosure for field use in harsh environments

Long Term: Multi-Channel

4 simultaneous voltage + 4 current channels. Cloud dashboard storing and comparing measurements over time

Machine Learning

ML model trained on thousands of circuit signatures for more accurate AI identification instead of groq AI

Commercial Product

Targeting engineering colleges, small workshops, field technicians across India

WiFi-CRO: Intelligent Measurement for Everyone

₹800 vs ₹50,000
Traditional oscilloscope cost comparison

16-Bit Resolution
62.5 microvolt precision

FFT + THD
Signal purity measurement

AI Powered
Circuit identification and health diagnosis

Trend Monitoring
Predictive fault detection

Any Device
Works on any browser — no software install

> **WiFi-CRO is not just a cheaper oscilloscope. It is a smarter one.**

By combining precise hardware measurement with AI intelligence, it makes professional circuit analysis accessible to every student, every technician, and every engineer — regardless of budget or location.